# MorpheOS — today

**Never breaks · Heals itself**

An operating system built on one promise — never breaks, and heals itself. That promise is **already real in a virtual machine**: it is measured, it is tested, and it is now graduating onto a real laptop.

This page is the results, in one read — what is real, what is measured, and what is still honest-to-goodness not built yet. The screenshots are the real launch screen ( `design/login.html` , the Three.js geomesh animation the machine boots), rendered in a browser at frozen timestamps.

## THE LAUNCH SCREEN




**Landed** — the geomesh resolves to matrix green, wordmark + tagline ( `?t=6.0s` )




**Assembling** — the mesh draws itself edge by edge, phasing colours ( `?t=2.0s` )




**First light** — early colour phase, rings orbiting ( `?t=0.5s` )

## THE WEDGE, MEASURED — NOT ESTIMATED

**~6–7 s**

Write a new system image into a slot and verify it by reading it back

the core of heal rung H4 — measured on the x86 test VM, 2026-10-11

**~2.4 s**

A boot reaches its health decision

measured on the x86 test VM, 2026-10-11

**0**

Wrong verdicts in the A/B trial-boot suite

good / unhealthy / crash scenarios all assert the exact expected boot sequence

## REAL NOW — VM-PROVEN

| WHAT | EVIDENCE |
|---|---|
| A/B boot — try once, commit on health, fall back automatically | good / unhealthy / crash scenarios, incl. a panicking kernel rolling back by boot counting |
| Update — write & verify a new system into the spare slot | `update` scenario: slot B boots the payload's marker |
| Heal — rebuild the failed slot from the running system | `heal` scenario (38 s), no 1.2 GiB copy, never the running slot |
| Transactions that roll themselves back | arm → apply → health-check → commit/rollback; a dead process rolls back, not hangs |
| Signed updates | payload opens only if a key in the image signed it (unsigned / wrong key / tampered refused) |
| Strata — versioned storage, rewind, ransomware detection | test-ransomware.sh: 24 files back byte-for-byte |
| Flow shell — a real Wayland compositor | move / resize / fullscreen / minimize / Alt+Tab on real client paths |
| Firefox on the shell, a page read off the screen | test-browser.sh reads the page's colours inside the VM |
| Wi-Fi picker — a WPA2 join typed on the keyboard | test-wifi.sh: wrong password refused, right one joins, DHCP logged |

| | |
|---|---|
| Recovery image, policy & the initramfs "way in" (S4) | merged #118–#120, 2026-10-18 |
| Bootable x86 USB image · Pi 5 image | x86 boots in QEMU; Pi builds and is read back (never booted) |

## NOT BUILT YET — THE HONEST PART

| WHAT | STATUS |
|---|---|
| Booted on **real hardware** | never, yet — the 2012 MacBook Pro native install is happening now |
| H4's immutable recovery image, H5 Phoenix restore | S4 recovery is in progress; Phoenix not started |
| `morphd` (PID 1) | its transaction half exists; the daemon does not |
| Capsules (Wine/Android), the AI layer, the mesh | documents, not code — deliberately re-sequenced behind the express road |

> **The boundary that matters:** "VM-proven" means a CI job boots it and reads the verdict off the disk. It does **not** mean it has run on a person's machine. Say "builds" or "VM-proven" — never "works" — until a person has sat at it.

## WHERE IT'S GOING

ADR 0014 ("express road first") re-sequenced the plan around one exit: **a general user boots a laptop, gets online, browses, and cannot make the machine unbootable.** The hard part — the unbreakable wedge — is already built and VM-proven; what's left on the express road is finishing the shell, then proving the whole stack on one real laptop. Every other dream (capsules, the AI layer, the mesh, Morph Link, fleet) stays in a widening backlog — **none of it dropped**, all of it gated behind a machine a person can actually use.

Source: github.com/robmitchellengineering-hub/MorpheOS · Apache-2.0 userland, GPL-2.0 kernel · main `8b31544` (2026-10-18) · numbers: x86 test VM, 2026-10-11, GitHub Actions run `38114088235` · screenshots: `design/login.html` rendered in Chrome (the repo's own launch screen).